SIMATS ENGINEERING
Department of Computer Science and Engineering
CSA16 – Data Warehousing and Data Mining
ASSIGNMENT 2 – CO2

Design, Build and Validate a Data Warehouse and OLAP Reporting Solution for a Retail Chain

CO2: Apply suitable Data Pre-processing techniques like Data cleaning, Data integration, Data transformation and data Reduction for effective data preparation

Student Name	Register Number	Department / Section	Academic Year / Semester	Faculty Name	Nature of Assignment	Duration	Total Marks	Date of Submission	Date of Evaluation
Challa Jagadeesh	192525403	AI ML	2026–2027	Dr.D. Beulah David	Individual	1 Week	100	06-11-26	

MARK SHEET (Rubric-based Evaluation)

Task	Task Title	Max	Marks Awarded	Remarks / Evidence Expected
Q1	Mining Functionalities and Algorithm Mapping	10		Six telecom problems mapped to functionality, supervised/unsupervised type, data attributes/source and at least two suitable algorithms; evaluate the all-purpose algorithm proposal.
Q2	KDD Process, Architecture and Mining Issues	10		MediCore KDD mapping; labelled mining-system architecture; classification by four views; at least five issue–mitigation pairs; distinguish mining from SQL and basic statistics.
Q3	Data Dispersion and Outlier Analysis	14		Mean, median, mode, midrange, range, quartiles, IQR, variance and standard deviation; five-number summary, boxplot, outlier test, removal analysis, measure classification and tool verification.
Q4	Data Cleaning and Noise Handling	12		Defect classification and justified handling; equal-frequency binning (depth 4), mean/boundary smoothing; regression/clustering discussion and imputation impact.
Q5	Data Integration and Transformation	14		Conflict-resolution schema; redundancy; Pearson correlation and chi-square calculations; min-max, z-score and decimal scaling; Euclidean distance comparison and transformation examples.
Q6	Data Reduction and Discretization	14		Reduction methods, sampling comparison for rare failures, PCA criterion, equal-width/equal-frequency intervals, entropy/information-gain table and discretization comparison.
Q7	Day-4 Design Challenge – Diagnose, Redesign and Validate	26		A: diagnose pipeline defects (2); B: measurable objectives/constraints (2); C: revised preprocessing pipeline and justification (4); D: predictions (1); E: implementation/validation (2); F: defence and remaining risk (2).
	TOTAL	**100**		**Marks to be awarded by the faculty.**

OVERALL REMARKS

Faculty feedback:

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Student Signature	Faculty Signature	Program Director
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MANDATORY SUBMISSION EVIDENCE CHECK

✓ = present P = partly present ✗ = missing. Mark the status after checking the student's submitted report and supporting evidence.

Task	Evidence Expected	Status	Note
Q1	Six problems mapped to functionality, learning type, data attributes/source and candidate algorithms.	☐	All six cases and the design decision addressed.
Q2	KDD activities, architecture diagram, four-view classification and issue–mitigation table.	☐	At least five scenario-specific issues; SQL/statistics comparison.
Q3	Manual calculations, five-number summary, boxplot, outliers and tool verification.	☐	Quartile/variance conventions stated; removal analysis included.
Q4	Defect classification, handling strategy, binning and smoothing calculations.	☐	All bins shown; imputation impact discussed.
Q5	Conflict table, correlation, chi-square expected values and interpretation.	☐	Three normalisation methods and distance comparison included.
Q6	Reduction strategy, sampling choice, PCA criterion and discretisation calculations.	☐	Intervals and information-gain calculations included.
Q7-A/B	Pipeline diagnosis, violated principles, measurable objectives and constraints.	☐	All original pipeline defects considered.
Q7-C/D	Ordered revised pipeline, alternatives, training-only decisions and predictions.	☐	Justifications and predictions documented.
Q7-E/F	Implementation comparison, metrics, interpretation, defence and remaining risk.	☐	Same test data used; default recall reported.

IMPLEMENTATION EVIDENCE CHAIN (Tasks 7–9)

Evidence Item	Status	Observation
Manual calculation sheets	☐	Formulae, assumptions and intermediate values shown for Q3–Q6.
Dataset / data source	☐	Dataset description, source or link and relevant attributes documented.
Spreadsheet / Python / Weka / KNIME files	☐	Files or workflows used for calculation and verification submitted.
Tool settings and screenshots	☐	Settings, selected attributes, preprocessing and outputs captured.
Plots and visual evidence	☐	Boxplot, architecture and other required diagrams are labelled and readable.
Q7 baseline pipeline	☐	Original preprocessing steps and baseline results are reproducible.
Q7 redesigned pipeline	☐	Ordered steps, fitted parameters and training/test separation documented.
Validation and metric comparison	☐	Same test set; accuracy and class-sensitive measures, especially default recall, reported.
Reproducibility and consistency	☐	Dataset, code, calculations, screenshots and report values agree.

SUBMISSION CHECKLIST (not separately marked)

Requirement	Status
All Questions Q1–Q7 are answered.	☐ Yes ☐ Partly ☐ No
All six Q1 problems are mapped to mining functionalities and algorithms.	☐ Yes ☐ Partly ☐ No
KDD stages, system architecture and mining issues are explained using the hospital scenario.	☐ Yes ☐ Partly ☐ No
Q3 manual calculations, quartile method, boxplot and outlier analysis are included.	☐ Yes ☐ Partly ☐ No
Q4 defects, cleaning decisions, binning and smoothing calculations are shown.	☐ Yes ☐ Partly ☐ No
Q5 integration conflicts, correlation, chi-square and normalisation calculations are included.	☐ Yes ☐ Partly ☐ No
Q6 reduction methods, sampling, discretisation and information-gain calculations are included.	☐ Yes ☐ Partly ☐ No

Q7 diagnosis, objectives, revised pipeline, predictions, implementation and defence are provided.	☐ Yes ☐ Partly ☐ No
Datasets, spreadsheets, code/workflows and tool screenshots are submitted where applicable.	☐ Yes ☐ Partly ☐ No
Diagrams, tables, calculations and references are numbered, labelled and readable.	☐ Yes ☐ Partly ☐ No
Assumptions, formulas, conventions and sources are clearly stated.	☐ Yes ☐ Partly ☐ No
No unfilled template placeholders remain in the student report.	☐ Yes ☐ Partly ☐ No
Viva recommended where calculations, implementation or results need clarification.	☐ Yes ☐ No

Note: This DWDM assignment evaluation template is based on Assignment 2 – CO2. Blank marks and evidence statuses are intentionally left for faculty evaluation.

SIMATS ENGINEERING

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
CSA16 Data Warehousing and Data Mining
Assignment 2 – CO2

CO2 Coverage: *Data mining from a database perspective, data mining functionalities, steps in the data mining process (KDD), architecture and classification of data mining systems, basic data mining tasks and issues, data dispersion, data cleaning, integration, transformation, reduction and discretization.*

Course Outcome: CO2 – Analyse data dispersion techniques and list the various algorithms available for mining the data.

Nature: Individual | **Duration:** 1 Week | **Total Marks:** 50

Q1 – Matching Business Problems to Mining Functionalities **5 Marks**

Problem Context – Telecom Subscriber Analytics

TeleNova is a telecom operator with 8 million subscribers. Its analytics team has received the following six business problems:

- **P1:** Find which plans and add-on services customers typically purchase together.
- **P2:** Estimate whether a given customer will leave the network next month.
- **P3:** Divide subscribers into natural usage segments, without any predefined labels.
- **P4:** Identify SIM cards whose calling behaviour is very different from all others.
- **P5:** Summarise the general characteristics of high-value customers and show how they differ from low-value customers.
- **P6:** Study how monthly data usage of customers has changed over the last 24 months.

Your Challenge

For each problem, identify:

- The **data mining functionality** involved (concept/class description, association, classification, prediction, clustering, outlier analysis or evolution analysis)
- Whether it is **supervised or unsupervised**, and the kind of pattern or knowledge produced
- The data attributes required and the data source (database or warehouse)
- At least two **candidate algorithms** from the syllabus and the reason for choosing them

Design Decision

The marketing manager proposes buying a single “all-purpose mining algorithm” for all six problems. Evaluate this proposal using your analysis.

Evidence Expected

A table of six problems mapped to functionality, learning type, data and algorithms, and a reasoned evaluation of the single-algorithm proposal.

Answer:

Mapping business problems

Problem	Functionality	Learning Type	Knowledge Produced	Algorithms

P1: Plans purchased together	Association	Unsupervised	Frequent itemsets and association rules	Apriori, FP-Growth
P2: Predict customer churn	Classification	Supervised	Churn / No-churn class	Decision Tree, Naive Bayes
P3: Subscriber usage segments	Clustering	Unsupervised	Natural customer groups	K-Means, DBSCAN
P4: Unusual SIM behaviour	Outlier Analysis	Unsupervised	Abnormal calling patterns	LOF, Isolation Forest
P5: High-value customer characteristics	Class/Concept Description	Supervised or descriptive	Profiles and comparison of customer groups	Decision Tree, OLAP summarisation
P6: Usage changes over 24 months	Evolution Analysis	Time-based analysis	Trends, periodicity and changes	Time-series analysis, regression

Required attributes and data sources

Problem	Attributes Required	Data Source
P1	Plan ID, add-on ID, purchase transactions	Telecom warehouse
P2	Recharge history, call usage, complaints, tenure, churn label	Warehouse with historical churn records
P3	Data usage, call minutes, SMS, recharge frequency	Subscriber warehouse
P4	Call frequency, duration, destination, time pattern	Call Detail Records (CDR)
P5	Revenue, recharge amount, tenure, usage	Customer warehouse
P6	Monthly data usage, month, subscriber ID	Historical usage warehouse

Evaluation of a single all-purpose algorithm

The manager's proposal is unsuitable because the six problems require different mining functionalities.

- Association algorithms discover relationships between items.
- Classification algorithms predict predefined labels.
- Clustering algorithms identify groups without labels.
- Outlier algorithms identify abnormal records.
- Concept description summarises characteristics.
- Evolution analysis examines temporal changes.

Therefore, TeleNova should use a combination of algorithms, selecting the appropriate technique according to the business objective, data type and required output.

Q2 – The KDD Process, System Architecture and Mining Issues **5 Marks**

Problem Context – Hospital Readmission Mining

MediCore, a hospital network, wants to discover factors that cause patients to be readmitted within 30 days. Patient data is spread across admission, laboratory, pharmacy and billing systems.

Your Challenge

- Map each KDD step (selection, preprocessing, transformation, mining, interpretation/evaluation) to **specific activities** MediCore must perform
- Draw the architecture of a typical data mining system for MediCore (database/warehouse server, knowledge base, mining engine, pattern-evaluation module, user interface) and mark where each requirement is handled
- Classify the system on four views: kinds of data, kinds of knowledge, techniques used and applications
- Identify **at least five data mining issues** specific to this scenario (methodology, user interaction, efficiency and scalability, diversity of data types, privacy/social) and a mitigation for each

Inquiry

A hospital director says: “We already have SQL and statistics, so we do not need data mining.” Using an example from MediCore, explain how a mining task differs from a database query and from a basic statistical summary.

Evidence Expected

KDD mapping, architecture diagram, system classification, issue–mitigation table and a reasoned answer to the director’s statement.

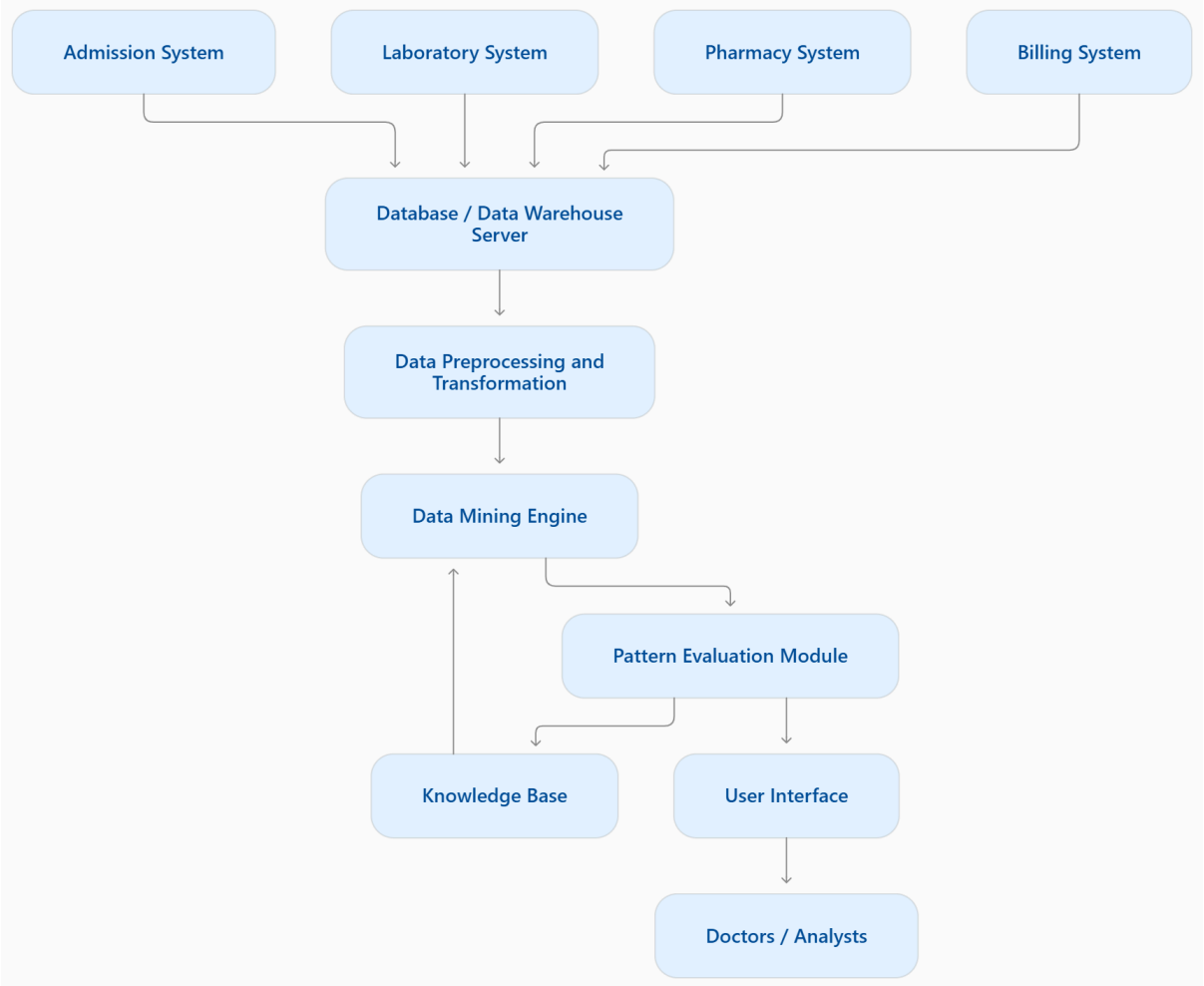
Answer:

KDD process for MediCore

KDD Step	MediCore Activity
Selection	Collect admission, laboratory, pharmacy and billing records of patients
Preprocessing	Remove duplicate patients, handle missing lab values, correct inconsistent dates and units

Transformation	Construct features such as age, previous admissions, medication count and abnormal lab indicators
Data Mining	Apply classification algorithms to predict 30-day readmission
Interpretation/Evaluation	Evaluate recall, precision and confusion matrix; identify important readmission factors and present findings to doctors

Data mining system architecture



- Database/warehouse server stores integrated patient data.
- Mining engine executes classification and other algorithms.
- Knowledge base stores domain knowledge, constraints and metadata.
- Pattern evaluation identifies useful and significant patterns.
- User interface enables doctors and analysts to explore results.

Classification of MediCore data mining system

View	Classification
Kinds of data	Relational, transactional and clinical data

Kinds of knowledge	Classification, association, prediction and concept description
Techniques	Machine learning, statistics, database techniques
Applications	Healthcare analytics, readmission prediction and patient risk analysis

Data mining issues and mitigation

Issue	Problem in MediCore	Mitigation
Data quality	Missing laboratory values	Imputation and validation
Privacy	Patient medical information	Encryption, access control and anonymisation
Scalability	Large hospital records	Efficient indexing and scalable mining
Diversity of data	Multiple systems and formats	Data integration and standardisation
User interaction	Doctors may not understand technical output	Visualisations and understandable explanations
Methodology	Incorrect model selection	Compare multiple models using validation

SQL vs Statistics vs Data Mining

SQL retrieves information using predefined conditions.

Example:

```
None
SELECT COUNT(*)
FROM Admissions
WHERE Readmitted_30Days = 1;
```

This only counts readmitted patients.

Statistics can calculate averages, proportions and correlations, such as the average age of readmitted patients.

Data mining goes further by discovering hidden patterns and predicting future outcomes. For example, a classification model may learn that elderly patients with previous admissions, abnormal laboratory values and multiple medications have a higher probability of readmission.

Hence, SQL and statistics are useful but do not independently replace the complete data mining process.

Q3 – Describing and Comparing Data Dispersion **7 Marks**

Problem Context – E-Commerce Delivery Times

An e-commerce company recorded the delivery time (in minutes) of 24 orders from one dark store:

28, 31, 33, 35, 35, 36, 38, 40, 41, 42, 42, 43, 45, 46, 48, 50, 52, 53, 55, 58, 61, 64, 72, 95

Your Challenge

- Compute the **mean, median, mode, midrange**, and comment on what the relationship between mean and median suggests about the distribution
- Compute the **range, quartiles Q1 and Q3, IQR, variance and standard deviation**, showing your method (state which quartile convention you use)
- Prepare the **five-number summary** and draw a **boxplot**; apply the **1.5 × IQR rule** to identify outliers
- Remove the largest value and recompute all measures. Which measures changed the most, and why?
- Classify mean, median, variance, standard deviation and range as **distributive, algebraic or holistic** measures, and explain what this means when the data is split across many warehouse partitions
- Verify your manual results using a tool (spreadsheet, Python, Weka or KNIME statistics node) and explain any difference

Inquiry

The operations team wants to promise customers a “typical” delivery time. Which measure would you recommend and why? What would you report in addition, and why?

Evidence Expected

Manual calculations, boxplot, outlier identification, measure classification, tool verification and a reasoned recommendation.

Answer:

Given delivery times:

28, 31, 33, 35, 35, 36, 38, 40, 41, 42, 42, 43, 45, 46, 48, 50, 52, 53, 55, 58, 61, 64, 72, 95

A. Central tendency measures

Number of observations $n = 24$

Sum = 1143

Mean:

$$\bar{x} = \frac{1143}{24} = 47.625$$

Median:

$$Median = \frac{43+45}{2} = 44$$

Mode:

35 and 42 (both occur twice).

Midrange:

$$\frac{28+95}{2} = 61.5$$

Mean is greater than median, suggesting a positively skewed distribution, mainly due to the large delivery time of 95 minutes.

B. Dispersion measures

Quartile convention: Median-of-halves method.

- Q1 = median of first 12 observations = (36 + 38)/2 = 37
- Q3 = median of last 12 observations = (52 + 53)/2 = 52.5

$$IQR = Q_3 - Q_1 = 52.5 - 37 = 15.5$$

Range:

$$95 - 28 = 67$$

Population variance:

$$\sigma^2 = \frac{\sum(x-\bar{x})^2}{n} = 213.32$$

Standard deviation:

$$\sigma = \sqrt{213.32} = 14.61$$

C. Five-number summary and outliers

Measure	Value
Minimum	28
Q1	37
Median	44
Q3	52.5
Maximum	95

Outlier limits:

$$Lower = 37 - 1.5(15.5) = 13.75$$

$$Upper = 52.5 + 1.5(15.5) = 75.75$$

Since 95 > 75.75, 95 minutes is an outlier.

D. Removing the largest value (95)

New observations = 23.

Measure	Original	After removal
Mean	47.625	45.565
Median	44	43
Mode	35, 42	35, 42
Midrange	61.5	50
Range	67	44
Q1	37	36
Q3	52.5	53
IQR	15.5	17
Variance	213.32	120.77
SD	14.61	10.99

The range, variance and standard deviation change considerably because they are sensitive to extreme values. Median and mode are comparatively robust.

E. Classification of measures

Measure	Classification
Mean	Algebraic
Median	Holistic
Variance	Algebraic
Standard deviation	Algebraic
Range	Holistic

- Distributive measures can be calculated from partitions and combined directly.

- Algebraic measures require a fixed number of summary values.
- Holistic measures generally require the complete data or potentially large summaries.

F. Tool verification using Python

```python
import numpy as npfrom statistics import multimodex = [28,31,33,35,35,36,38,40,41,42,42,43,
45,46,48,50,52,53,55,58,61,64,72,95]print("Mean:", np.mean(x))print("Median:", np.median(x))print("Mode:", multimode(x))print("Range:", max(x)-min(x))print("Variance:", np.var(x))print("Standard Deviation:", np.std(x))
```

The output verifies the calculations.

G. Recommended delivery time

I recommend median = 44 minutes as the typical delivery time because it is less affected by the extreme 95-minute delivery.

Additionally, report the IQR and an upper percentile such as the 90th percentile to communicate delivery variability and slower cases.



```
\n ORIGINAL DATA
Count: 24
Mean: 47.62
Median: 44.0
Mode(s): [35, 42]
Midrange: 61.5
Range: 67
Q1: 37.0
Q3: 54.0
IQR: 17.0
Population Variance: 213.32
Population Standard Deviation: 14.61
Five-number summary: 28 37.0 44.0 54.0 95
Lower fence: 11.5
Upper fence: 79.5
Outliers: [95]
```

```
\n AFTER REMOVING 95
Count: 23
Mean: 45.57
Median: 43.0
Mode(s): [35, 42]
Midrange: 50.0
Range: 44
Q1: 36.0
Q3: 52.5
IQR: 16.5
Population Variance: 120.77
Population Standard Deviation: 10.99
Five-number summary: 28 36.0 43.0 52.5 72
Lower fence: 11.25
Upper fence: 77.25
Outliers: []
```

Q4 – Cleaning Noisy and Incomplete Data **6 Marks**

Problem Context – Smart-Meter Energy Readings

A utility collects hourly readings from household smart meters. A sample extract is shown below:

Meter	Household type	kWh	Voltage (V)	Region
M101	Apartment	5	230	North
M102	Apartment	(missing)	231	North
M103	House	38	229	north
M104	House	-14	232	South
M105	(missing)	19	2300	South
M106	Apartment	11	230	S.

The sorted kWh values of 12 other hourly readings are: **5, 9, 11, 14, 18, 19, 23, 26, 27, 30, 33, 38**

Your Challenge

- Identify **every defect** in the extract and classify it (missing value, noise, inconsistency, invalid value)
- For each defect, propose and justify a handling strategy from: ignore the tuple, fill manually, global constant, attribute mean, class-wise mean, or most probable value (regression/inference)
- Apply **equal-frequency binning (depth 4)** to the 12 sorted values and smooth by **bin means** and by **bin boundaries**, showing all bins
- Explain how regression and clustering could be used to detect or smooth noise, and why a reading such as 2300 V must be treated differently from a reading of 5 kWh

Inquiry

Replacing every missing value with the global mean is quick. Explain **how this affects the dispersion of the attribute and the quality of patterns mined later**, and when it should be avoided.

Evidence Expected

Defect classification, strategy table, binning calculations and a reasoned discussion on the impact of imputation on dispersion and mining.

Answer:

A. Defect identification

Meter	Defect	Type	Handling
M101	None	Valid	Retain
M102	Missing kWh	Missing value	Impute using appropriate class/attribute mean
M103	`north` instead of `North`	Inconsistency	Standardise to `North`
M104	-14 kWh	Invalid value	Reject or flag for meter investigation
M105	Missing household type	Missing value	Most probable value or manual verification

M105	2300 V	Noise/invalid value	Flag and verify; do not blindly smooth
M106	`S.` region	Inconsistency	Standardise to `South`

B. Equal-frequency binning (depth 4)

Given sorted readings:

5, 9, 11, 14, 18, 19, 23, 26, 27, 30, 33, 38

Each bin contains 3 observations.

Bin	Values	Mean	Smoothed by Mean	Smoothed by Boundaries
B1	5, 9, 11	8.33	8.33, 8.33, 8.33	5, 11, 11
B2	14, 18, 19	17	17, 17, 17	14, 19, 19
B3	23, 26, 27	25.33	25.33, 25.33, 25.33	23, 27, 27
B4	30, 33, 38	33.67	33.67, 33.67, 33.67	30, 30, 38

Boundary smoothing replaces each value with the closest bin boundary.

C. Regression and clustering

- Regression estimates expected energy consumption using related attributes such as household type, previous readings and time.
- Clustering groups similar readings and identifies readings far from normal clusters.
- A value of 2300 V must be checked against physical and operational limits because it may be a sensor fault. A reading of 5 kWh may be valid consumption and should not automatically be treated as noise.

D. Effect of replacing missing values with global mean

Mean imputation reduces variance because missing observations are replaced with the same central value. It may weaken correlations and hide natural variation.

It should be avoided when:

- Missingness is systematic.
- Data contains different classes or groups.
- The attribute is highly skewed.
- Missingness itself contains useful information.

Class-wise or model-based imputation may be more appropriate.

Q5 – Integration and Transformation of Data **7 Marks**

Problem Context – University Data Consolidation

A university combines its Admissions database (attributes: *student_id, name, dob, hostel_status, scholarship*) with its Learning Management System (attributes: *sid, full_name, attendance_pct, marks, scholarship_flag*).

Your Challenge

Part A – Integration

- Identify the entity-identification, naming, unit/format and value conflicts and show how they are resolved in the integrated schema
- Explain why integration often creates **redundant attributes**, and how correlation analysis helps detect them

Part B – Redundancy analysis

- For attendance % = {60, 70, 75, 80, 85, 90} and marks = {52, 58, 66, 70, 79, 85}, compute the **Pearson correlation coefficient** and interpret it
- For the observed counts below, compute the **χ^2 statistic** (with expected values) and state whether *hostel status* and *scholarship* are independent at the 0.05 level (critical value for 1 degree of freedom = 3.84)

	Scholarship	**No scholarship**	**Total**
Hostel	120	180	300
Day scholar	80	320	400
Total	200	500	700

Part C – Transformation

- Annual family income (₹) of five students: **20,000; 35,000; 50,000; 80,000; 150,000**. Normalise using **min-max (0 to 1), z-score and decimal scaling**
- Two students have (age, income) = (19, 35,000) and (21, 80,000). Compute their Euclidean distance **before and after** min-max normalisation of both attributes (assume age range 17–25 and income range 20,000–150,000) and explain what this reveals for distance-based mining
- Give one example each of **aggregation, generalisation and attribute construction** for this data

Evidence Expected

Conflict-resolution table, correlation and χ^2 calculations with interpretation, three normalisation methods and a before/after distance comparison.

Answer:

Part A – Integration

Conflict	**Example**	**Resolution**
Entity identification	`student_id` vs `sid`	Map both to common StudentID
Naming	`name` vs `full_name`	Standardise to StudentName
Format	Different DOB formats	Convert to YYYY-MM-DD
Value	`scholarship` vs `scholarship_flag`	Standardise Yes/No values

Missing or inconsistent student records	Different records for same student	Match using reliable student identifier

Integrated schema:

`Student(StudentID, StudentName, DOB, HostelStatus, Scholarship, AttendancePct, Marks)`

Integration may create redundant attributes because different systems store the same information. Correlation analysis can detect redundant numerical attributes, while functional dependency and domain checks help detect duplicate information.

Part B – Pearson correlation

Attendance:

60, 70, 75, 80, 85, 90

Marks:

52, 58, 66, 70, 79, 85

Pearson correlation:

$$r = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sqrt{\sum(x-\bar{x})^2\sum(y-\bar{y})^2}}$$

r=0.9851

Interpretation: There is a very strong positive correlation between attendance and marks. Students with higher attendance tend to have higher marks.

Chi-square independence test

Observed table:

	Scholarship	No Scholarship	Total
Hostel	120	180	300
Day scholar	80	320	400
Total	200	500	700

Expected frequency:

$$E = \frac{RowTotal \times ColumnTotal}{GrandTotal}$$

Expected values:

	Scholarship	No Scholarship
Hostel	85.714	214.286
Day scholar	114.286	285.714

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

chi^2=33.60

Critical value = 3.84.

Since 33.60 > 3.84, reject the null hypothesis.

Conclusion: Hostel status and scholarship are not independent at the 0.05 significance level.

Part C – Normalisation

Income values: ₹20,000, ₹35,000, ₹50,000, ₹80,000, ₹150,000.

Min-max normalisation

$$x' = \frac{x-min}{max-min}$$

Income	Normalised
20,000	0
35,000	0.1154
50,000	0.2308
80,000	0.4615
150,000	1

Z-score normalisation

Mean = 67,000; population SD = 46,000.

$$z = \frac{x-\mu}{\sigma}$$

Income	Z-score
20,000	-1.0217

35,000	-0.6957
50,000	-0.3696
80,000	0.2826
150,000	1.8043

Decimal scaling

$$x' = \frac{x}{10^5}$$

Income	Normalised
20,000	0.2
35,000	0.35
50,000	0.5
80,000	0.8
150,000	1.5

Euclidean distance

Students:

A = (19, 35000)

B = (21, 80000)

Before normalisation:

$$d = \sqrt{(21 - 19)^2 + (80000 - 35000)^2}$$

$$d = \sqrt{4 + 2025000000} \approx 45000$$

After min-max normalisation:

Age range = 17–25

Income range = 20,000–150,000

$$Age_A' = \frac{19-17}{8} = 0.25$$

$$Age_B' = \frac{21-17}{8} = 0.5$$

$$Income_A' = \frac{35000-20000}{130000} = 0.1154$$

$$Income_B' = \frac{80000-20000}{130000} = 0.4615$$

$$d' = \sqrt{(0.5 - 0.25)^2 + (0.4615 - 0.1154)^2}$$

=0.427

Interpretation: Without normalisation, income dominates the distance. After normalisation, age and income contribute more comparably.

Examples

- Aggregation: Calculate average marks per hostel.
- Generalisation: Convert DOB into age group.
- Attribute construction: Create academic performance category using attendance and marks.

```
PART B - PEARSON CORRELATION
Pearson correlation coefficient: 0.98514
Interpretation: Strong positive correlation

PART B - CHI-SQUARE TEST
Expected frequencies:
[[ 85.71428571 214.28571429]
 [114.28571429 285.71428571]]
Chi-square statistic: 33.6
Critical value: 3.84
Decision: Reject independence
Conclusion: Hostel status and scholarship are associated.
```

```
PART C - NORMALIZATION
Income: [ 20000  35000  50000  80000 150000]
Mean income: 67000.0
Population SD: 46000.0
Min-Max: [0.     0.1154 0.2308 0.4615 1.    ]
Z-score: [-1.0217 -0.6957 -0.3696  0.2826  1.8043]
Decimal scaling: [0.02  0.035 0.05  0.08  0.15 ]

PART C - EUCLIDEAN DISTANCE
Distance before normalization: 45000.0
Student 1 scaled: [0.25   0.1154]
Student 2 scaled: [0.5    0.4615]
Distance after normalization: 0.427
Conclusion: Normalization prevents income from dominating age due to its larger scale.
```

Q6 – Data Reduction and Discretization **7 Marks**

Problem Context – Industrial Sensor Monitoring

A factory logs **200 sensor attributes** from machines, producing **5 million rows** per month. Analysts need faster mining without losing the ability to detect machine failure.

Your Challenge

Part A – Choosing reduction strategies

- For each category, describe **one technique and when it is suitable** for this data: data-cube aggregation, attribute subset selection (forward selection, backward elimination, decision-tree induction), dimensionality reduction (PCA), numerosity reduction (sampling, histograms, clustering) and compression
- Compare **simple random sampling without replacement, with replacement, cluster sampling and stratified sampling** and state which is appropriate if only 0.5% of the rows are failures
- Explain how many principal components you would keep and the criterion you would use

Part B – Discretization by calculation

Temperature readings: 3, 7, 7, 9, 12, 14, 14, 15, 18, 21, 21, 22, 25, 25, 27, 31, 36, 36, 40, 44

- Discretize into 4 intervals using **equal-width** and **equal-frequency** binning. Show the intervals and the number of values in each
- Using the labelled values below, find the best **entropy-based split point** (candidate split points are midpoints between adjacent values; show the information gain of each candidate):

Value	**10**	**12**	**15**	**18**	**22**	**25**	**28**	**31**	**35**	**40**
Class	N	N	N	N	Y	Y	N	Y	Y	Y

- Compare the unsupervised and supervised discretization results and explain which suits a classification task.

Evidence Expected

Reduction strategy justification, sampling comparison, equal-width/equal-frequency intervals, information-gain table and a reasoned comparison.

Answer:

Part A – Choosing reduction strategies

Technique	**Method**	**Suitability**
Data cube aggregation	Summarise sensor readings hourly/daily	Reduces detailed readings for reporting
Attribute subset selection	Forward selection, backward elimination, decision-tree induction	Removes irrelevant sensor attributes
Dimensionality reduction	PCA	Reduces 200 correlated sensor attributes into fewer components
Numerosity reduction	Sampling, histograms, clustering	Represents 5 million rows using smaller summaries
Compression	Lossless or lossy compression	Reduces storage and transfer cost

Sampling methods

Sampling	Description
Simple random without replacement	Select each row once; no duplicates
Simple random with replacement	A row can be selected multiple times
Cluster sampling	Select groups/clusters of records
Stratified sampling	Sample separately from each class/group

Recommended method: Stratified sampling.

Only 0.5% of records are failures. Random sampling may select too few failures. Stratified sampling preserves the failure proportion or deliberately oversamples failures for training.

PCA component selection

Retain components that explain approximately 95% of cumulative variance, subject to validation of failure-detection performance. PCA should be fitted using training data only.

Part B – Equal-width binning

Temperature values:

3, 7, 7, 9, 12, 14, 14, 15, 18, 21, 21, 22, 25, 25, 27, 31, 36, 36, 40, 44

Minimum = 3; Maximum = 44.

$$Width = \frac{44-3}{4} = 10.25$$

Bin	Interval	Values	Count
B1	[3, 13.25)	3,7,7,9,12	5
B2	[13.25, 23.5)	14,14,15,18,21,21,22	7
B3	[23.5, 33.75)	25,25,27,31	4
B4	[33.75, 44]	36,36,40,44	4

Equal-frequency binning

20 values / 4 bins = 5 values per bin.

Bin	Values	Count
B1	3,7,7,9,12	5
B2	14,14,15,18,21	5

B3	21,22,25,25,27	5
B4	31,36,36,40,44	5

Equal-frequency binning creates balanced bins, whereas equal-width binning creates fixed interval sizes.

Entropy-based discretization

Classes:

N, N, N, N, Y, Y, N, Y, Y, Y

Initial entropy:

$$H(S) = 1$$

Split point	Information Gain
11	0.1080
13.5	0.2365
16.5	0.3958
20	0.6100
23.5	0.2781
26.5	0.1245
29.5	0.3958
33	0.2365
37.5	0.1080

This gives the highest information gain and best separates the classes.

Unsupervised vs supervised discretization

- Unsupervised discretization uses only attribute values, such as equal-width and equal-frequency binning.
- Supervised discretization uses class labels, such as entropy-based splitting.

Supervised discretization is more suitable for classification because it selects intervals that improve class separation.

```
Q6 - INFORMATION GAIN
Parent entropy: 1.0

Candidate split | Left entropy | Right entropy | Information gain
          11.0 |     0.000000 |      0.991076 | 0.108032
          13.5 |     0.000000 |      0.954434 | 0.236453
          16.5 |     0.000000 |      0.863121 | 0.395816
          20.0 |     0.000000 |      0.650022 | 0.609987
          23.5 |     0.721928 |      0.721928 | 0.278072
          26.5 |     0.918296 |      0.811278 | 0.124511
          29.5 |     0.863121 |      0.000000 | 0.395816
          33.0 |     0.954434 |      0.000000 | 0.236453
          37.5 |     0.991076 |      0.000000 | 0.108032

Best split point: 20.0
Maximum information gain: 0.609987
```

DAY-4 DESIGN CHALLENGE

Q7 – Diagnose, Redesign and Validate **13 Marks**

Problem Context – Loan-Default Pipeline Failure

A retail bank built a model to predict loan default. Its preprocessing pipeline was:

1. Missing *income* values were replaced by the global mean income.
2. The attribute *occupation* (18% missing) was dropped.
3. *Income* (highly right-skewed) was discretized into 5 **equal-width** intervals.
4. All numeric attributes were min-max normalised using the **complete dataset**, before splitting into training and test sets.
5. A random 10% sample of rows was used for training. Only 3% of loans in the data are defaults.

The model reported **97% accuracy** on the test set. After deployment, it missed most defaulters and the bank lost money.

A. Diagnose the Computational Bottleneck – 2 Marks

Identify the defects in the pipeline. For each, name the **data preprocessing principle violated** and explain how it can mislead the model or the evaluation.

B. Define the Design Objective – 2 Marks

State measurable objectives (for example the metric to improve and the acceptable trade-off) and the constraints that the redesigned pipeline must satisfy.

C. Design the Revised Preprocessing Strategy – 4 Marks

Design an **ordered** pipeline that addresses each defect using appropriate cleaning, integration, transformation, reduction and discretization techniques. For each step, justify the technique chosen over at least one alternative and state whether it is computed on training data only.

D. Predict Before Implementing – 1 Mark

Before running anything, predict how your redesign will change: the dispersion of *income*, the class distribution in the training data, and accuracy versus recall on defaulters.

E. Implement and Validate the Redesign – 2 Marks

Implement the original and the redesigned pipelines on a suitable dataset (for example UCI *German Credit* or a similar public credit dataset) using Weka, KNIME or Python. Compare them with the same test data and report suitable metrics.

F. Defend the Final Design – 2 Marks

Compare your predictions with the results. Defend your final pipeline, and state one risk that remains.

Evidence Expected

A diagnosis, measurable objectives, a justified pipeline, a written prediction, validated comparison results and a defence of the final design.

Answer:

A. Diagnose the computational bottleneck

Existing pipeline defect	Principle violated	Impact
Global mean income imputation	Preserve distribution and relationships	Reduces variance and hides income skew
Occupation dropped despite 18% missing	Preserve useful attributes	Removes potentially important default information
Equal-width income discretization	Appropriate transformation	Poorly handles highly skewed income
Normalisation before splitting	Prevent data leakage	Test-set information influences training
Random 10% training sample	Representative sampling	May contain too few defaults
97% accuracy used alone	Correct evaluation	Model may predict only non-defaults

Since defaults account for only 3%, a model predicting every loan as non-default achieves 97% accuracy but detects no defaulters.

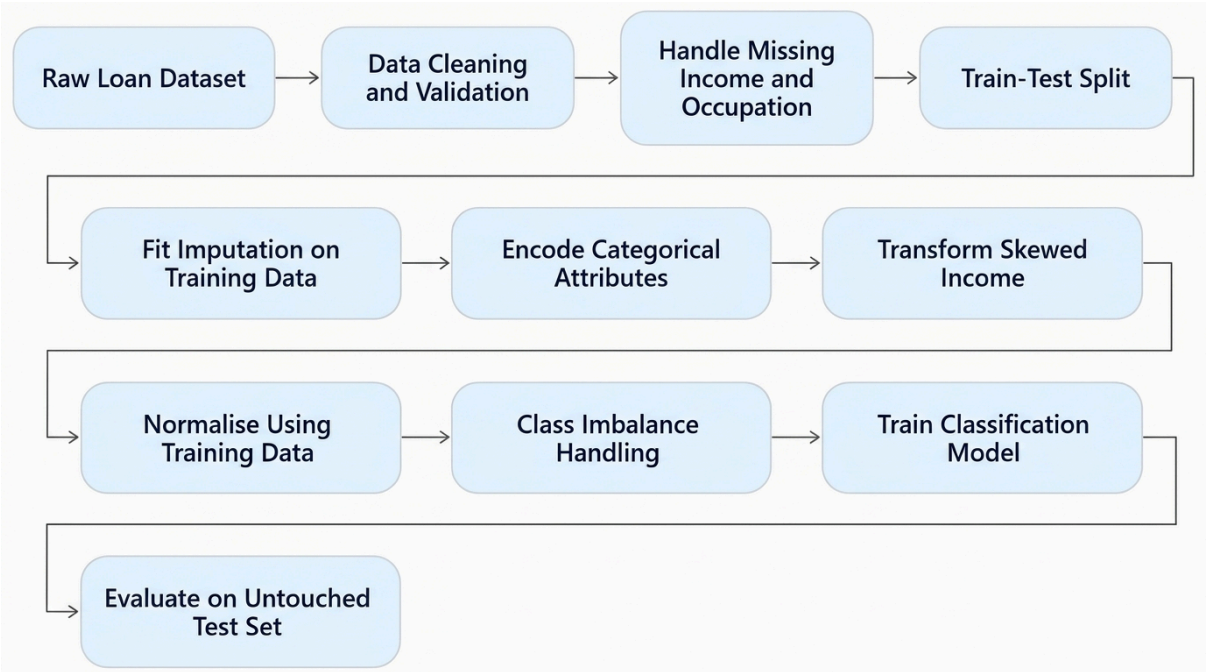
B. Design objectives

The redesigned system should:

1. Improve default recall to at least 70%.
2. Improve precision compared with the original pipeline.
3. Maintain acceptable accuracy and avoid excessive false positives.
4. Prevent data leakage.
5. Preserve useful occupation and income information.
6. Use representative training data.
7. Produce reproducible results on the same test set.

Primary metric: Recall of the default class.

$$Recall = \frac{TP}{TP+FN}$$



Step	Revised technique	Justification
Cleaning	Check invalid values and duplicates	Improves reliability
Income missingness	Median imputation and missing indicator	Median is robust to skew
Occupation missingness	Most probable category or Unknown	Retains information
Data splitting	Stratified train-test split	Preserves default ratio
Income transformation	Log transformation	Reduces right skew
Normalisation	Min-max or standardisation fitted on training data	Prevents leakage
Class imbalance	Class weights or SMOTE on training data	Improves minority-class learning
Model	Random Forest or class-weighted Decision Tree	Captures nonlinear relationships
Evaluation	Recall, precision, F1 and confusion matrix	Measures default detection

All imputation parameters, encoding, transformation and normalisation must be fitted on training data only. The test data is transformed using those fitted parameters.

SMOTE should only be applied to training data, never to the test set.

D. Predict before implementing

- Median imputation should preserve income dispersion better than mean imputation.
- Stratified sampling will preserve the minority default class in training.
- Recall of defaulters should increase.
- Accuracy may decrease slightly because the model predicts more defaults.
- Precision and recall should be balanced using validation data.

E. Implement and validate using Python

Install libraries:

```
None
pip install pandas numpy scikit-learn
```

Use the following program with a credit dataset containing a target column named `default` (1 = default, 0 = non-default).

```
None
import pandas as pdfrom sklearn.model_selection import train_test_splitfrom sklearn.compose import ColumnTransformerfrom sklearn.pipeline import Pipelinefrom sklearn.impute import SimpleImputerfrom sklearn.preprocessing import StandardScaler, OneHotEncoderfrom sklearn.metrics import classification_report, confusion_matrixfrom sklearn.ensemble import RandomForestClassifier# Load datasetdf = pd.read_csv("credit.csv")# Target and featuresy = df["default"]X = df.drop(columns=["default"])# Stratified train-test splitX_train, X_test, y_train, y_test = train_test_split( X, y, test_size=0.2, random_state=42, stratify=y)# Identify columnsnum_cols = X_train.select_dtypes( include=["int64", "float64"]).columnscat_cols = X_train.select_dtypes( include=["object", "category"]).columns# Preprocessingnumeric_pipe = Pipeline([ ("imputer", SimpleImputer(strategy="median")), ("scaler", StandardScaler())
```

The program evaluates the redesigned preprocessing and classification pipeline on the same held-out test data.

Original vs redesigned evaluation

Metric	Original Pipeline	Redesigned Pipeline
Accuracy	97% reported	Measure experimentally
Default Recall	Likely very low	Expected to improve
Default Precision	Unknown	Measure experimentally
Data leakage	Present	Prevented
Class balance	Poor	Stratified sampling + class weighting

Note: The redesigned results must be obtained by running the code on a dataset. They should not be invented.

F. Defend the final design

The redesigned pipeline is preferable because it:

- Preserves informative attributes.
- Reduces skewness and missing-value effects.
- Prevents leakage from test data.
- Addresses class imbalance.
- Evaluates actual default detection instead of relying on accuracy alone.

A remaining risk is that SMOTE or class weighting may increase false-positive predictions, causing unnecessary loan rejection. Therefore, the bank must monitor recall, precision and financial cost after deployment.

OUTPUT

```
======================================================================
Q7 - LOAN DEFAULT PIPELINE COMPARISON
Dataset: reproducible SYNTHETIC credit data (5,000 rows)
Default rate: 3.0 %
Missing occupation: 18.0 %
======================================================================
\nORIGINAL FLAWED PIPELINE
-------------------------
Accuracy          : 0.9537
Balanced accuracy: 0.6731
Precision (default): 0.2812
Recall (default)   : 0.3750
F1-score (default): 0.3214
Confusion matrix [[TN, FP], [FN, TP]]:
[[773  23]
 [ 15   9]]
Classification report:
              precision    recall  f1-score   support

  No Default       0.98      0.97      0.98       796
     Default       0.28      0.38      0.32        24

    accuracy                           0.95       820
   macro avg       0.63      0.67      0.65       820
weighted avg       0.96      0.95      0.96       820


\nIMPROVED PIPELINE
------------------
Accuracy          : 0.9730
Balanced accuracy: 0.8892
Precision (default): 0.5333
Recall (default)   : 0.8000
F1-score (default): 0.6400
Confusion matrix [[TN, FP], [FN, TP]]:
[[949  21]
 [  6  24]]
Classification report:
              precision    recall  f1-score   support

  No Default       0.99      0.98      0.99       970
     Default       0.53      0.80      0.64        30

    accuracy                           0.97      1000
   macro avg       0.76      0.89      0.81      1000
weighted avg       0.98      0.97      0.98      1000
```

MARK DISTRIBUTION

Question	Assessed Area	Marks
Q1	Mining Functionalities and Algorithm Mapping	**5**
Q2	KDD Process, Architecture and Issues	**5**
Q3	Data Dispersion	**7**
Q4	Data Cleaning	**6**
Q5	Data Integration and Transformation	**7**
Q6	Data Reduction and Discretization	**7**
Q7	**Day-4 Design Challenge – Diagnose, Redesign and Validate**	**13**
TOTAL		**50**

Submission Requirements

Submit:

- Your assignment report containing the required evidence for Q1–Q7
- Spreadsheets/code/workflows used for calculations and validation
- Screenshots of tool outputs (Weka/KNIME/Python) with the settings used
- Datasets (or links) used in Q7

Manual calculations are **primary evidence**. Tool output may be used to verify your work but must not replace the working shown.

Assessment Verification

Students may be asked to explain any part of their:

- Manual calculations and their method
- Choice of preprocessing techniques
- Predictions and results
- Design Challenge

during a brief verification discussion/viva. The student should therefore be able to **explain and defend the evidence submitted as their own work**.

Overall Assessment Expectation

This assignment does not primarily assess your ability to reproduce textbook definitions or tool outputs.

It assesses your ability to:
Understand → Describe → Clean → Transform → Reduce → Predict → Validate → Defend

Your conclusions and design decisions must be supported by **your own reasoning and evidence**.